

AGRIVISION

AI-Powered Pre-Symptomatic Crop Intelligence System *Detects Stress Before It's Visible. Act Before Damage Is Done.*

1. Problem Statement

India's 140M+ small and marginal farmers collectively lose ₹50,000–₹1,00,000 crore annually to preventable crop losses — not because treatments don't exist, but because detection always comes too late. Crop stress (nutrient deficiency, water stress, pest infestation) is identified only after visible symptoms appear, by which point ~20% damage is already irreversible. Existing precision-agriculture tools work at the macro scale and miss early physiological changes. Compounding this, 65% of Indian farmland lacks reliable internet, making cloud-only solutions inaccessible where they matter most.

2. Proposed Solution

AgriVision is an AI-powered, connectivity-independent crop intelligence system that uses **hyperspectral imaging + multi-sensor fusion** to detect crop stress at **~0.1% severity** weeks before any visible symptom appears.

Parameter	Existing Solutions	AgriVision
Detection Stage	Post-symptomatic	Pre-symptomatic (~0.1% stress)
Sensing Method	RGB cameras / visual	Hyperspectral + soil + weather fusion
Connectivity	Requires internet	Fully offline with SMS fallback
AI Inference	Cloud-only	Edge AI + cloud sync
Output	Raw data/dashboards	Actionable vernacular recommendations

3. System Architecture

Sensing Layer → A drone equipped with a hyperspectral camera captures field data periodically, reducing cost and enabling large-area monitoring in a single pass. Raw data is **preprocessed and synchronized**, and key vegetation indices (NDVI, NDRE, OSAVI, SGI) are extracted using **MATLAB**.

In the **edge processing layer**, fused data is analyzed using a lightweight CNN model to detect crop stress type and severity. A **recommendation engine** generates precise, zone-specific actions like irrigation scheduling and fertilizer application. Insights are delivered via a **dual system**—mobile app (online) and SMS alerts, LoRa communication (offline), ensuring accessibility for farmers across varying connectivity conditions.

4. Five-Dimensional Crop Intelligence Model

AgriVision analyzes crops across five integrated parameters: **Nutritional Health** (NPK detection), **Water Stress**, **Disease Detection**, **Environmental Stress**, and an **Overall Crop Health Index** updated every 30 minutes from sensor fusion data.

Environmental	Social	Economic
30–40% reduction in pesticide & fertiliser overuse	Empowers small farmers with expert-level insights	15–20% yield improvement per season
20–30% water savings via precision irrigation	Vernacular SMS reduces digital literacy barrier	₹8,000–15,000 input cost savings per farmer/year
Reduced chemical runoff → healthier soil	Aligns with PM-KISAN & digital agriculture goals	SaaS + API licensing for startup viability

6. Why AgriVision Wins

- **Pre-symptomatic detection at ~0.1% stress**: no existing affordable product does this for small Indian farmers.
- **Offline-first edge AI**: full inference runs on-device with zero internet dependency.
- **Hyperspectral + multi-sensor fusion** builds a 3D crop health model across spectral, biochemical, and climatic dimensions.
- **Vernacular SMS output** bridges the last mile: farmers receive advice in Hindi/Marathi, not raw data.
- **Built for Bharat**: ₹10K prototype cost, GSM fallback, FPO-ready shared deployment (1 device per 10–15 acres).

Startup Potential: India's precision agriculture market is projected at \$1.2B by 2027. Proprietary Indian crop spectral datasets + offline edge AI model create a strong IP moat. Eligible for DST, NABARD, and ICAR grants.

AgriVision combines hyperspectral science, edge AI, drone sensing, and LoRa/GSM communication to deliver predictive, scalable, and farmer-friendly crop intelligence — designed not for boardrooms, but for the ground realities of Bharat.